

Diffusion-Sparse Dual-Prior Extended Split Gibbs Sampling for SAR-GMTI

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Abstract—Detecting and estimating ground moving targets in heterogeneous SAR scenes is challenging due to strong clutter and complex terrain. This paper proposes an extended split Gibbs sampling framework that jointly leverages diffusion-based priors for static clutter and sparsity priors for moving targets. By decoupling likelihood and prior terms with auxiliary variables, the method enables simultaneous clutter reconstruction and target parameter estimation within a unified Bayesian framework. The approach effectively captures heterogeneous clutter structures and accurately recovers target positions and velocities, demonstrating superior performance compared with conventional techniques.

Index Terms—SAR-GMTI, split Gibbs sampling, diffusion model, compressed sensing, Langevin dynamics.

I. INTRODUCTION

Synthetic aperture radar (SAR) achieves high-resolution imaging of static scenes by synthesizing a large aperture through the motion of the platform [1]. However, the conventional SAR imaging model inherently assumes stationary scatterers, which leads to position displacement and defocusing for moving targets [2]. In addition, strong ground clutter and its associated Doppler broadening can significantly degrade the signal-to-clutter-plus-noise ratio of moving targets, making slow-moving targets particularly susceptible to being masked by background clutter and noise [3]. As a result, the detection, localization, and imaging of moving targets become considerably more challenging, presenting significant obstacles for SAR-based ground moving target indication (GMTI).

In SAR-GMTI, the dual objectives are the detection of moving targets and the estimation of their parameters, including velocity and position [4]. This work focuses on the parameter estimation stage. Existing methods can be broadly categorized according to how prior information is exploited. Traditional signal processing approaches primarily rely on idealized motion models and physical signatures [5]–[7], without explicitly incorporating structural priors on the signals, rendering these methods vulnerable to strong clutter. Compressed sensing (CS) techniques partially mitigate this issue by enforcing sparsity constraints on moving targets [8], [9], yet they often neglect the structure of background clutter. To further enhance target–clutter separation, some studies introduce

additional low-rank priors for static clutter [10]. However, such handcrafted priors exhibit limited expressive power in highly heterogeneous environments, such as mountainous terrains, where rapid terrain elevation variations and complex dielectric properties [4], [11] cannot be well represented by simple low-rank models, resulting in degraded performance in moving target separation and parameter estimation.

Recently, diffusion models [12] have demonstrated remarkable capabilities in generative modeling by learning complex, high-dimensional data distributions, enabling a more accurate characterization of heterogeneous clutter in SAR scenarios. Leveraging their superior expressive power, diffusion models are particularly well-suited for capturing the intricate geometric structures and electromagnetic scattering properties inherent in nonstationary environments. Furthermore, diffusion-based plug-and-play (PnP) frameworks have been shown to converge asymptotically to the true posterior distribution [13], [14], providing a principled means to incorporate learned priors in inverse problems. In our previous work [15], diffusion models were successfully applied to SAR image reconstruction, demonstrating their effectiveness in capturing domain-specific priors for static scenes. Building upon this foundation, accurate modeling of static clutter can effectively suppress target obscuration caused by clutter, thereby enabling robust moving target extraction and precise parameter estimation in SAR-GMTI.

In this paper, we propose an extended split Gibbs sampling framework for SAR-GMTI, in which static clutter and moving targets are decoupled and jointly recovered within a unified reconstruction process. Specifically, a pretrained diffusion model is employed as a powerful prior for static clutter, eliminating the need for explicit pre-processing or handcrafted low-rank constraints, while a sparsity prior is imposed on the moving target component. This framework enables fine-grained reconstruction of static clutter alongside simultaneous recovery of moving target positions and velocities, obviating the need for secondary parameter estimation or relocation procedures. Experimental results demonstrate that the proposed approach achieves superior performance in complex, heterogeneous environments, substantially enhancing both imaging fidelity and estimation accuracy.

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II. BACKGROUND AND PROBLEM FORMULATION

A. Signal Model for SAR-GMTI

Consider an L -channel SAR system flying at constant altitude H along the x -axis with velocity v , transmitting an LFM waveform. The inter-element spacing is d , with only the first channel transmitting and all channels receiving echoes.

For a moving target at $(x_0, y_0, 0)$ with velocity (v_x, v_y) , the equivalent monostatic slant range for the l -th channel can be approximated by a second-order Taylor expansion [3]:

$$R(\eta, l) \approx R_0 + \frac{v_x x_0}{R_0} \frac{ld}{2v} + \frac{v_x x_0 - (v - v_y)y_0}{R_0} \eta + \frac{(v - v_y)^2}{2R_0} \eta^2,$$

with $R_0 = \sqrt{x_0^2 + y_0^2 + H^2}$ and slow time η .

After down conversion and dechirp, the received signal is

$$y(\tau, \eta, l) = \alpha \exp(-j2\pi \frac{2K_r}{c} R_0 \tau) \exp(-j2\pi \frac{d}{\lambda v} v_r l) \exp(j2\pi (\frac{K_a}{v} y_0 - \frac{2}{\lambda} v_r) \eta), \quad (1)$$

where α denotes the back-scattering coefficient, τ is the fast time, and $v_r = \frac{v_x x_0}{R_0}$ is the radial velocity of the target. K_r and K_a denote the range and azimuth modulation rates, respectively, and λ is the radar wavelength. For static clutter, $v_r = 0$, no azimuth shift or inter-channel phase occurs.

By sampling the fast time and slow time as $\tau = n/f_s$ and $\eta = mT_r$, respectively, and discretizing the observed scene into a three-dimensional grid with P , Q , and K points along the range, azimuth, and spatial frequency dimensions, the discrete received echoes from moving targets and static clutter can be expressed as

$$y_m[n, m, l] = \sum_{p,q,k}^{P,Q,K} \alpha_{p,q,k} \exp(j2\pi(f_{az}[q] - f_{as}[k])m) \exp(-j2\pi f_{rg}[p]n) \exp(-j2\pi f_{sp}[k]l),$$

$$y_c[n, m, l] = \sum_{p,q}^{P,Q} \alpha_{p,q} \exp(-j2\pi f_{rg}[p]n) \exp(j2\pi f_{az}[q]m),$$

where $y_m[n, m, l]$ and $y_c[n, m, l]$ denote the discrete echoes from moving targets and static clutter, respectively; $\alpha_{p,q,k}$ and $\alpha_{p,q}$ are the corresponding back-scattering coefficients. $f_{rg} = \frac{2K_r}{cf_s} R_0$, $f_{az} = \frac{K_a T_r}{v} y_0$, and $f_{sp} = \frac{d}{\lambda v} v_r$ are range, azimuth, and spatial frequency, respectively. $f_{as} = \frac{2T_r}{\lambda} v_r$ represents the azimuth frequency shift induced by target motion.

Let $\mathbf{s} \in \mathbb{C}^{P \times Q \times K}$ and $\mathbf{x} \in \mathbb{C}^{P \times Q}$ denote the moving target and clutter coefficients. The overall observation with noise $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$ is

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathcal{B}(\mathbf{s}) + \mathbf{n}, \quad (2)$$

where $\mathcal{A}(\cdot) : \mathbb{C}^{P \times Q} \rightarrow \mathbb{C}^{N \times M \times L}$ and $\mathcal{B}(\cdot) : \mathbb{C}^{P \times Q \times K} \rightarrow \mathbb{C}^{N \times M \times L}$ denote the measurement operators corresponding to static clutter and moving targets, respectively.

B. MAP-based Reconstruction

Exploiting the sparsity of moving targets and the statistical structure of static clutter, the joint reconstruction of $\mathbf{s}$ and $\mathbf{x}$ is formulated within a maximum a posteriori (MAP) framework. Given the observation model in (2), the MAP estimation problem is defined as

$$\begin{aligned} (\hat{\mathbf{x}}, \hat{\mathbf{s}}) &= \arg \max_{\mathbf{x}, \mathbf{s}} p(\mathbf{x}, \mathbf{s} | \mathbf{y}) \\ &= \arg \max_{\mathbf{x}, \mathbf{s}} p(\mathbf{y} | \mathbf{x}, \mathbf{s}) p(\mathbf{x}) p(\mathbf{s}) \\ &= \arg \min_{\mathbf{x}, \mathbf{s}} \mathcal{L}(\mathbf{x}, \mathbf{s}; \mathbf{y}) + \phi(\mathbf{x}) + \psi(\mathbf{s}), \end{aligned} \quad (3)$$

where the likelihood function satisfies $p(\mathbf{y} | \mathbf{x}, \mathbf{s}) \propto \exp(-\mathcal{L}(\mathbf{x}, \mathbf{s}; \mathbf{y}))$, and the negative log-likelihood is defined as $\mathcal{L}(\mathbf{x}, \mathbf{s}; \mathbf{y}) = \frac{1}{\sigma^2} \|\mathbf{y} - \mathcal{A}(\mathbf{x}) - \mathcal{B}(\mathbf{s})\|_2^2$. To promote sparse representations of moving targets, a Laplacian prior is imposed on $\mathbf{s}$, while $\mathbf{x}$ is regularized using a structural prior that captures the statistical properties of static clutter:

$$p(\mathbf{s}) \propto \exp(-\psi(\mathbf{s})), \quad p(\mathbf{x}) \propto \exp(-\phi(\mathbf{x})), \quad (4)$$

where $\psi(\mathbf{s}) = \mu \|\mathbf{s}\|_1$ with μ controlling the sparsity level, and $\phi(\mathbf{x})$ encodes the prior knowledge of static clutter.

To effectively characterize the complex statistics of heterogeneous clutter, we adopt a diffusion-model-based prior for the clutter component. Diffusion models are capable of learning the score function s_θ of high-dimensional data distributions directly from samples, without requiring an explicit analytical form of the underlying probability density function [15]. Specifically, the learned score network approximates the gradient of the log-prior as $s_\theta(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$.

However, due to the implicit and parameterized nature of the diffusion prior, the MAP optimization problem in (3) does not admit a closed-form solution. To circumvent this difficulty, we instead perform posterior sampling from the distribution $\exp(-\mathcal{L}(\mathbf{x}, \mathbf{s}; \mathbf{y})) p(\mathbf{x}) p(\mathbf{s})$, which can be interpreted as a *stochastic* and *soft* relaxation of the MAP estimation.

III. EXTENDED SPLIT GIBBS SAMPLING BASED ON DIFFUSION-SPARSE DUAL-PRIOR

A. Extended Split Gibbs Sampling Framework

To efficiently address the MAP inference problem with implicit and non-analytical priors, we adopt an *extended split Gibbs sampling* (ESGS) framework. The key idea is to introduce auxiliary variables $(\mathbf{u}, \mathbf{v})$ to decouple the data-fidelity and prior terms, yielding the following augmented joint posterior distribution:

$$p(\mathbf{x}, \mathbf{s}, \mathbf{u}, \mathbf{v} | \mathbf{y}) \propto \exp\{-\mathcal{L}(\mathbf{u}, \mathbf{v}; \mathbf{y}) - \phi(\mathbf{x}) - \mu \|\mathbf{s}\|_1 - \frac{1}{2\rho^2} (\|\mathbf{x} - \mathbf{u}\|_2^2 + \|\mathbf{s} - \mathbf{v}\|_2^2)\}, \quad (5)$$

where $\rho > 0$ is a quadratic penalty parameter that controls the strength of coupling between the auxiliary variables and the original unknowns.

At iteration κ , Gibbs sampling proceeds by sequentially drawing samples from the following conditional distributions:

1) **Likelihood update for u (u -step):**

$$\mathbf{u}^{(\kappa+1)} \propto \exp \left\{ -\mathcal{L}(\mathbf{u}, \mathbf{v}^{(\kappa)}; \mathbf{y}) - \frac{1}{2\rho^2} \|\mathbf{x}^{(\kappa)} - \mathbf{u}\|_2^2 \right\}. \quad (6)$$

2) **Likelihood update for v (v -step):**

$$\mathbf{v}^{(\kappa+1)} \propto \exp \left\{ -\mathcal{L}(\mathbf{u}^{(\kappa+1)}, \mathbf{v}; \mathbf{y}) - \frac{1}{2\rho^2} \|\mathbf{s}^{(\kappa)} - \mathbf{v}\|_2^2 \right\}. \quad (7)$$

3) **Diffusion prior update (x -step):**

$$\mathbf{x}^{(\kappa+1)} \propto \exp \left\{ -\phi(\mathbf{x}) - \frac{1}{2\rho^2} \|\mathbf{x} - \mathbf{u}^{(\kappa+1)}\|_2^2 \right\}. \quad (8)$$

4) **Sparse prior update (s -step):**

$$\mathbf{s}^{(\kappa+1)} \propto \exp \left\{ -\mu \|\mathbf{s}\|_1 - \frac{1}{2\rho^2} \|\mathbf{s} - \mathbf{v}^{(\kappa+1)}\|_2^2 \right\}. \quad (9)$$

From an optimization perspective, ESGS can be interpreted as a stochastic analogue of half-quadratic splitting (HQS) method [14], where each conditional sampling step corresponds to a gradient descent update perturbed by stochasticity. This interpretation naturally accommodates implicit priors, such as diffusion models, within a unified Bayesian inference framework.

B. Likelihood Updates

The likelihood updates in (6) and (7) can be implemented using the *Unadjusted Langevin Algorithm* (ULA). Specifically, the continuous-time Langevin dynamics for $\mathbf{u}_t$ and $\mathbf{v}_t$ are given by

$$d\mathbf{u}_t = \left[-\nabla_{\mathbf{u}} \mathcal{L}(\mathbf{u}_t, \mathbf{v}; \mathbf{y}) - \frac{1}{\rho^2} (\mathbf{u}_t - \mathbf{x}) \right] dt + \sqrt{2} d\mathbf{B}_t, \quad (10)$$

$$d\mathbf{v}_t = \left[-\nabla_{\mathbf{v}} \mathcal{L}(\mathbf{u}, \mathbf{v}_t; \mathbf{y}) - \frac{1}{\rho^2} (\mathbf{v}_t - \mathbf{s}) \right] dt + \sqrt{2} d\mathbf{B}_t, \quad (11)$$

where $\mathbf{B}_t$ denotes standard Brownian motion, and the quadratic penalty terms enforce consistency with the current estimates $(\mathbf{x}, \mathbf{s})$.

In practice, these stochastic differential equations are discretized using the Euler–Maruyama scheme. Let γ_c and γ_m denote the step sizes for clutter and moving-target updates, respectively. The discrete-time iterations at step i are then given by

$$\mathbf{u}_{i+1} = \mathbf{u}_i - \gamma_c \nabla_{\mathbf{u}} \mathcal{L}(\mathbf{u}_i, \mathbf{v}; \mathbf{y}) - \frac{\gamma_c}{\rho^2} (\mathbf{u}_i - \mathbf{x}) + \sqrt{2\gamma_c} \boldsymbol{\epsilon}_i, \quad (12)$$

$$\mathbf{v}_{i+1} = \mathbf{v}_i - \gamma_m \nabla_{\mathbf{v}} \mathcal{L}(\mathbf{u}, \mathbf{v}_i; \mathbf{y}) - \frac{\gamma_m}{\rho^2} (\mathbf{v}_i - \mathbf{s}) + \sqrt{2\gamma_m} \boldsymbol{\epsilon}_i, \quad (13)$$

where $\boldsymbol{\epsilon}_i \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ represents standard Gaussian noise.

C. Diffusion Prior Update

Following [14], for the $\mathbf{x}$ -step, we employ a diffusion-based generative model (called EDM) [16]. In EDM, the forward noising process is formulated as

$$p(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{s}(t)\mathbf{x}_0, s^2(t)\sigma^2(t)\mathbf{I}), \quad (14)$$

where $s(t)$ and $\sigma(t)$ denote the scaling and noise schedules, respectively. By Bayes' theorem, the posterior distribution of $\mathbf{x}_0$ conditioned on $\mathbf{x}_t$ can be expressed as

$$p(\mathbf{x}_0 | \mathbf{x}_t) \propto p(\mathbf{x}_t | \mathbf{x}_0) p(\mathbf{x}_0) \propto \exp \left\{ -\phi(\mathbf{x}_0) - \frac{1}{2\sigma^2(t)} \|\mathbf{x}_0 - \frac{\mathbf{x}_t}{s(t)}\|_2^2 \right\}. \quad (15)$$

Comparing (15) with the $\mathbf{x}$ -step conditional in (8), we establish a direct correspondence by matching the diffusion noise level with the quadratic penalty, i.e., selecting t^* such that $\sigma(t^*) = \rho$. Accordingly, the reverse diffusion process is initialized from $\mathbf{x}_{t^*} = s(t^*)\mathbf{u}$, which ensures consistency with the auxiliary variable in the ESGS framework.

The reverse diffusion sampling can be implemented either through the reverse-time stochastic differential equation (SDE) or the probability flow ordinary differential equation (PF-ODE). In this work, we adopt the PF-ODE formulation for improved sampling efficiency, given by

$$d\mathbf{x}_t = \left[\frac{\dot{s}(t)}{s(t)} \mathbf{x}_t - s^2(t) \dot{\sigma}(t) \sigma(t) \nabla_{\mathbf{x}_t} \log p \left(\frac{\mathbf{x}_t}{s(t)}; \sigma(t) \right) \right] dt, \quad (16)$$

where $\dot{s}(t)$ and $\dot{\sigma}(t)$ denote time derivatives. The score function is approximated using a pre-trained denoising network $D_\theta(\cdot)$ as

$$\nabla_{\mathbf{x}_t} \log p \left(\frac{\mathbf{x}_t}{s(t)}; \sigma(t) \right) \approx \frac{D_\theta(\mathbf{x}_t; \sigma(t)) - \mathbf{x}_t}{\sigma^2(t)}. \quad (17)$$

In practice, the PF-ODE in (16) is discretized over a sequence of time steps $\{t_i\}_{i=i^*}^{N-1}$, yielding the numerical update

$$\begin{aligned} \mathbf{x}_{i+1} = \mathbf{x}_i + (t_{i+1} - t_i) & \left[\left(\frac{\dot{\sigma}(t_i)}{\sigma(t_i)} + \frac{\dot{s}(t_i)}{s(t_i)} \right) \mathbf{x}_i \right. \\ & \left. - \frac{\dot{\sigma}(t_i)s(t_i)}{\sigma(t_i)} D_\theta \left(\frac{\mathbf{x}_i}{s(t_i)}; \sigma(t_i) \right) \right]. \end{aligned} \quad (18)$$

D. Sparse Prior Update

In the $\mathbf{s}$ -step, the objective is to draw samples of $\mathbf{s}$ from the conditional distribution in (9). Owing to the non-differentiability of the ℓ_1 norm, conventional Langevin dynamics cannot be directly applied. To overcome this limitation, we adopt the *Proximal Langevin Algorithm* (PxLA) [17], which naturally incorporates non-smooth priors through a proximal operator.

Specifically, the smooth quadratic term is handled via Langevin dynamics, while the non-smooth sparsity-promoting prior is enforced by a proximal operator. This results in the following iterative PxLA update:

$$\mathbf{s}_{i+1} = \text{prox}_{\psi}^{\delta} \left(\mathbf{s}_i - \frac{\delta}{\rho^2} (\mathbf{s}_i - \mathbf{v}) + \sqrt{2\delta} \boldsymbol{\epsilon}_i \right). \quad (19)$$

where δ denotes the step size. For the ℓ_1 -norm prior $\psi(\mathbf{s}) = \mu \|\mathbf{s}\|_1$, the proximal operator admits a closed-form solution given by the element-wise soft-thresholding function:

$$\text{prox}_{\mu \|\cdot\|_1}^{\delta}(\mathbf{z}) = \mathcal{S}_{\mu\delta}(\mathbf{z}) = \mathbf{z} \odot \left(1 - \frac{\mu\delta}{|\mathbf{z}|} \right)_+.$$

IV. EXPERIMENTS

Datasets. Experiments were conducted on mountainous Sentinel-1 SAR scenes exhibiting strong heterogeneity. Two moving targets were synthetically embedded at range-azimuth indices (20, 50) and (200, 100) with radial velocities of -5 m/s and 12 m/s, respectively.

Comparison Methods. The proposed method was compared with three representative baselines: 1) *FFT*, which exploits no prior information; 2) *LASSO*, which enforces only

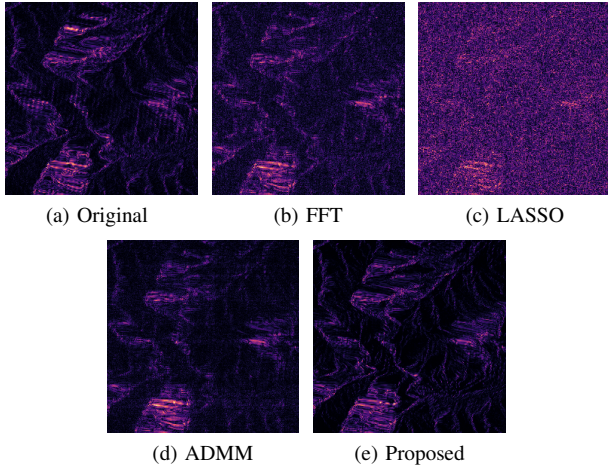


Fig. 1: Qualitative results of static clutter.

target sparsity; 3) *Low-rank ADMM*, which combines a low-rank clutter prior with a sparse target prior.

Evaluation Metrics. Clutter reconstruction was evaluated using NMSE and SSIM, while moving target recovery was quantified by the F1-score, reflecting the accuracy of target localization and velocity estimation.

Parameter Settings. The penalty parameter was scheduled as $\rho_k = \max\{\beta^k \rho_0, \rho_{\min}\}$ with $\rho_0 = 1$, $\rho_{\min} = 0.1$, and $\beta = 0.9$. Step sizes were set to $\gamma_c = 6 \times 10^{-4}$, $\gamma_m = 5 \times 10^{-4}$, and $\delta = 1 \times 10^{-3}$. Each sampling stage employed 100 inner iterations, and a total of 50 Gibbs iterations were performed.

Experimental results. Under the conditions of SCR = -30 dB and CNR = 5 dB, static clutter reconstruction was first evaluated. As illustrated in Fig. 1, the proposed method preserves significantly richer structural details and texture information of the original scene compared with the competing approaches. In contrast, FFT and LASSO suffer from severe noise contamination and loss of fine structures, while the low-rank ADMM method tends to oversmooth local details due to the restrictive low-rank constraint. Quantitative results are summarized in Table I. The proposed approach achieves the lowest NMSE and the highest SSIM among all methods, confirming its superior capability in accurately modeling and reconstructing heterogeneous clutter.

TABLE I: Quantitative metrics of results.

Method	FFT	LASSO	ADMM	Proposed
NMSE (dB) ↓	0.801	3.356	-2.047	-2.113
SSIM ↑	0.3772	0.1454	0.4163	0.5621
F1 ↑	—	0.0006	0.4016	1.0000
Clock time (s)	—	~ 99	~ 140	~ 109

For moving target recovery, azimuth slices were extracted for visual inspection, as shown in Fig. 2. The proposed method and ADMM exhibit sharply focused peaks at the correct target azimuth locations, whereas FFT and LASSO produce severely defocused results, characterized by a broadened main lobe and strong sidelobe contamination from residual clutter and noise. The quantitative evaluation using F1 scores, reported in Table I, further substantiates these observations. The proposed method attains a perfect F1 score of 1.0, accurately recovering

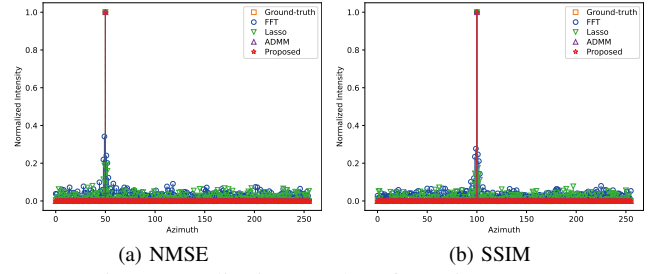


Fig. 2: Qualitative results of moving targets.

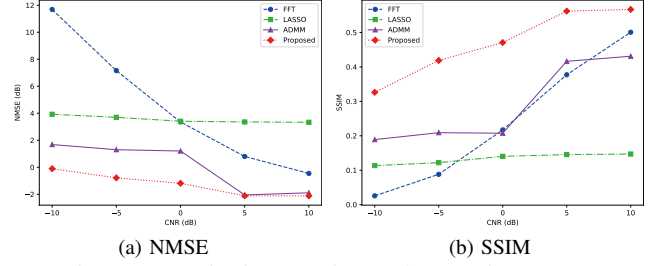


Fig. 3: Quantitative metrics under varying CNRs.

both the target locations and radial velocities without false alarms. In contrast, the low-rank ADMM method achieves an F1 score of 0.4016, indicating the presence of substantial false detections, despite correctly identifying the true targets.

In terms of computational complexity, the proposed method has a higher per-iteration cost (2.18 s/iter) due to multiple update steps, compared to 0.012 s/iter and 0.04 s/iter for LASSO and ADMM, respectively. However, it converges within tens of iterations, yielding a total runtime on the order of 10^2 seconds, while LASSO and ADMM fail to achieve comparable accuracy even after thousands of iterations. Moreover, traditional methods are sensitive to step size selection, and increasing the step size often causes numerical instability or divergence (NaN), limiting their robustness in SAR-GMTI scenarios. Overall, the proposed method achieves superior reconstruction performance with comparable or lower total computational cost.

To further evaluate robustness under different noise conditions, the clutter-to-noise ratio (CNR) was varied over $\{-10, -5, 0, 5, 10\}$ dB. As shown in Fig. 3, the proposed method consistently achieves the lowest NMSE and highest SSIM across all noise levels, demonstrating strong robustness and stability in severely noisy and heterogeneous clutter environments.

V. CONCLUSION

In this paper, an extended split Gibbs sampling framework is proposed for SAR-GMTI by integrating diffusion-based plug-and-play priors with sparsity priors. The proposed method enables the joint reconstruction of heterogeneous static clutter and accurate estimation of moving target parameters within a unified Bayesian framework. Experimental results demonstrate that the proposed approach consistently outperforms conventional methods in both clutter recovery and target estimation, especially in complex heterogeneous environments.

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